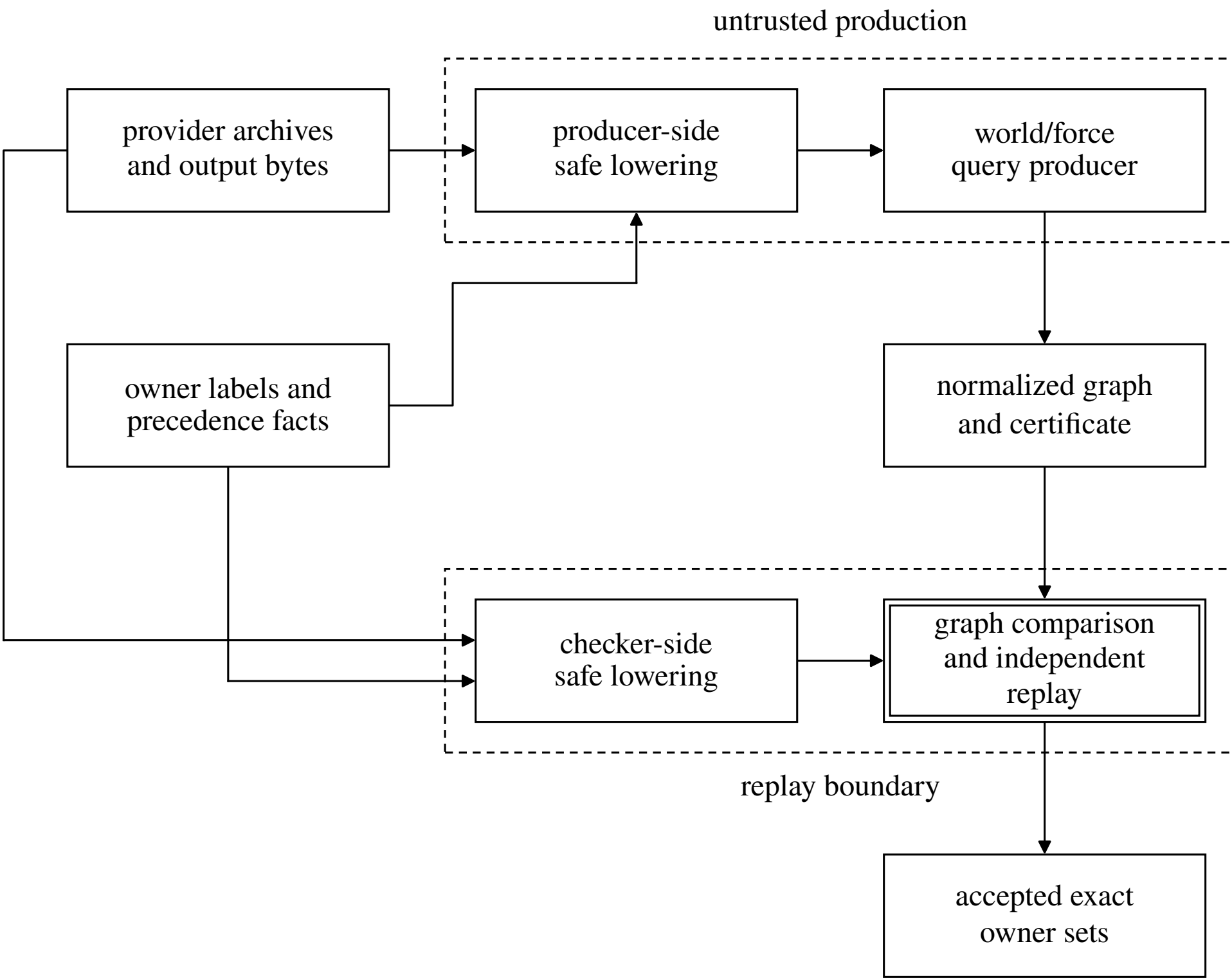
untrusted production
provider archives and output bytes
producer-side safe lowering
world/force query producer
owner labels and precedence facts
normalized graph and certificate
checker-side safe lowering
graph comparison and independent replay
replay boundary
accepted exact owner sets